

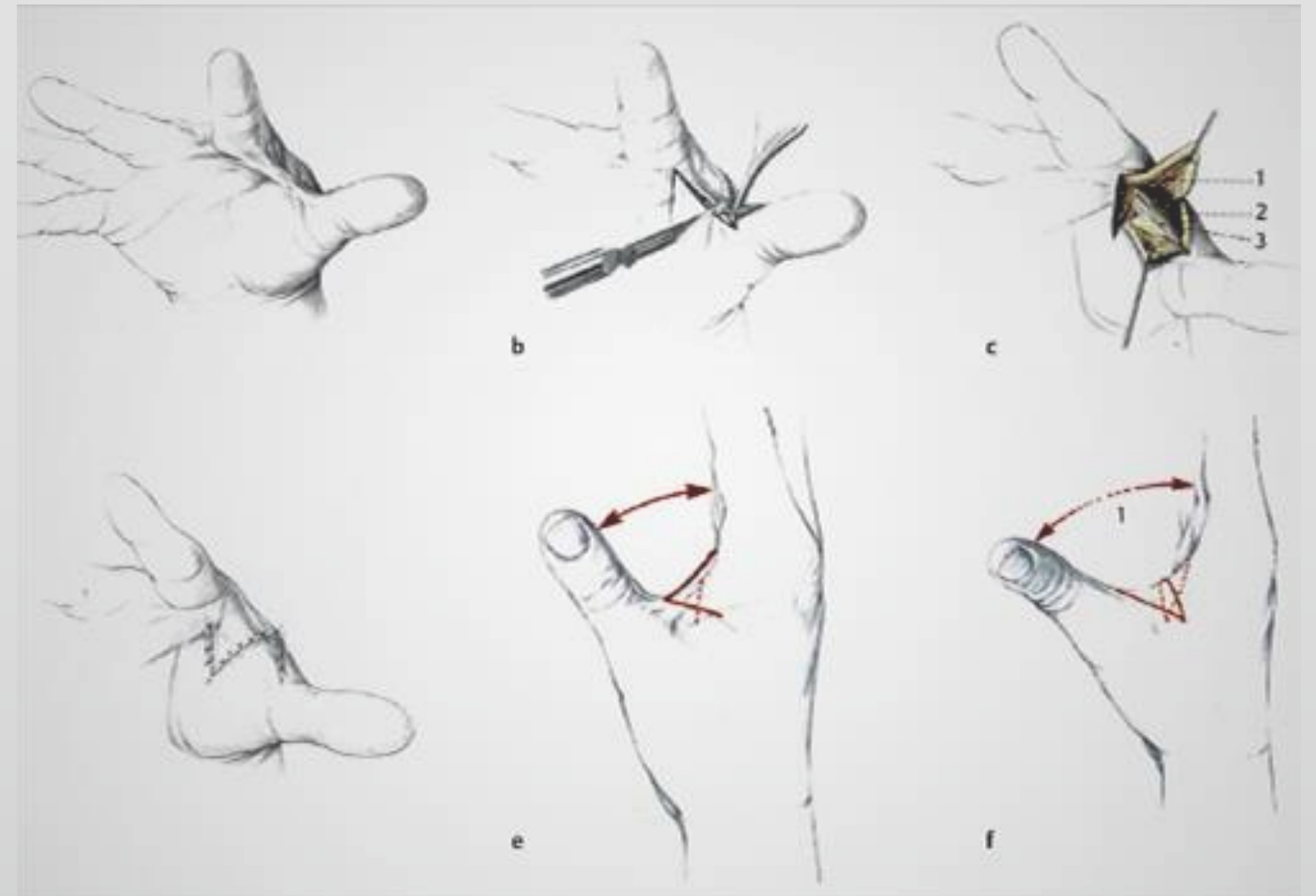


Z-

RECONSTRUCTIVE SURGERY GUIDE

PLASTY

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<https://plasticsurgerykey.com/77-z-plasty/>

SECTION 1

INTRODUCTION

INDICATIONS | ANATOMY

INDICATIONS

- Lengthen a scar
- Reorient a scar

Why?

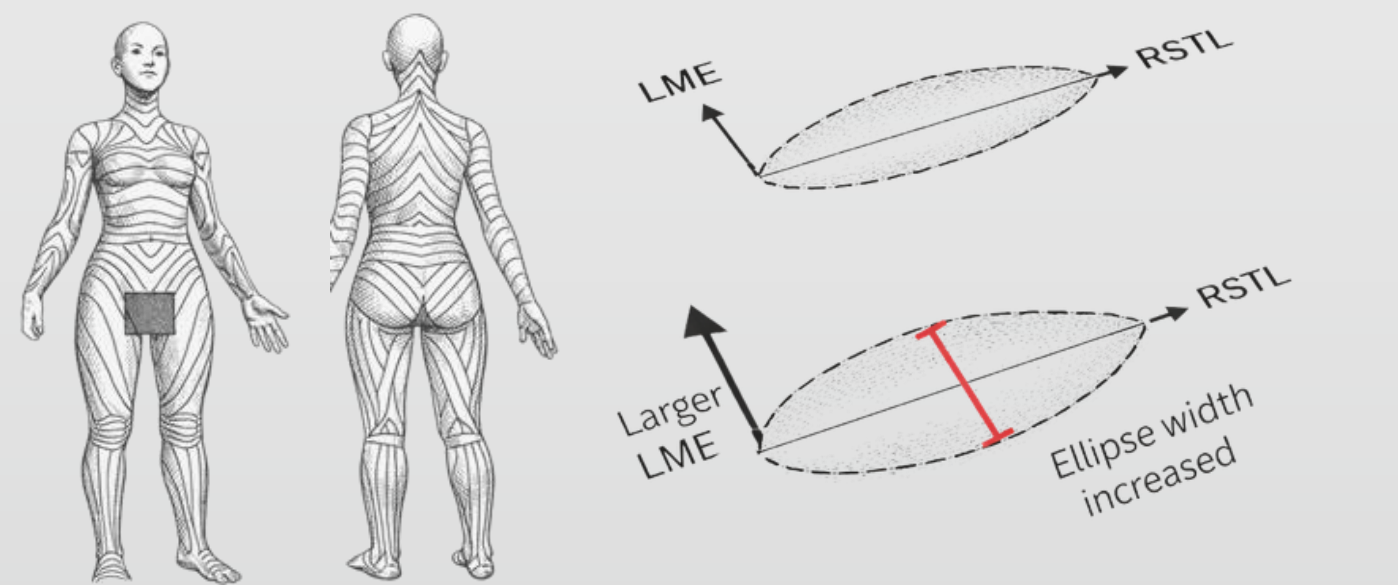
Scars and contractures can restrict movement, especially single band contractures.

LINES OF TENSION

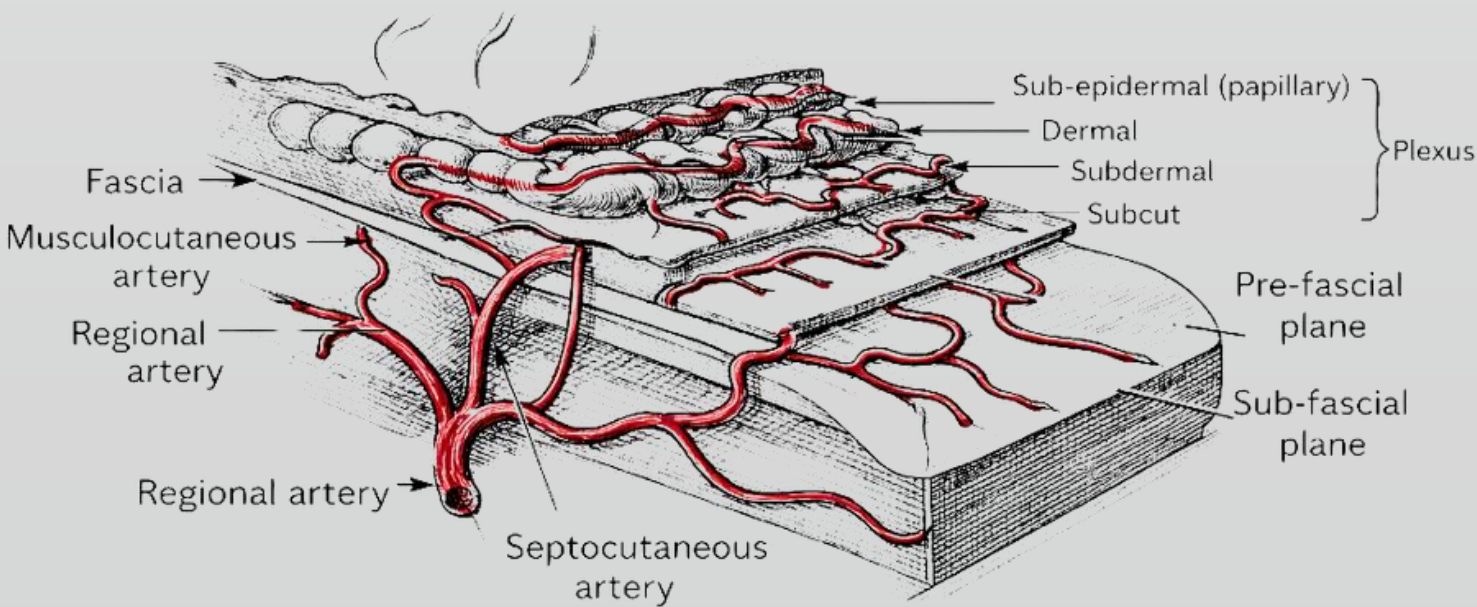
LME = Lines of Maximum Extensibility
RSTL = Relaxed Skin Tension Lines

These form an ellipse (length along RSTL, width along LME).

• **Key point:** New scars should align with the RSTL.



MICROVASCULAR ANATOMY



DESIGN

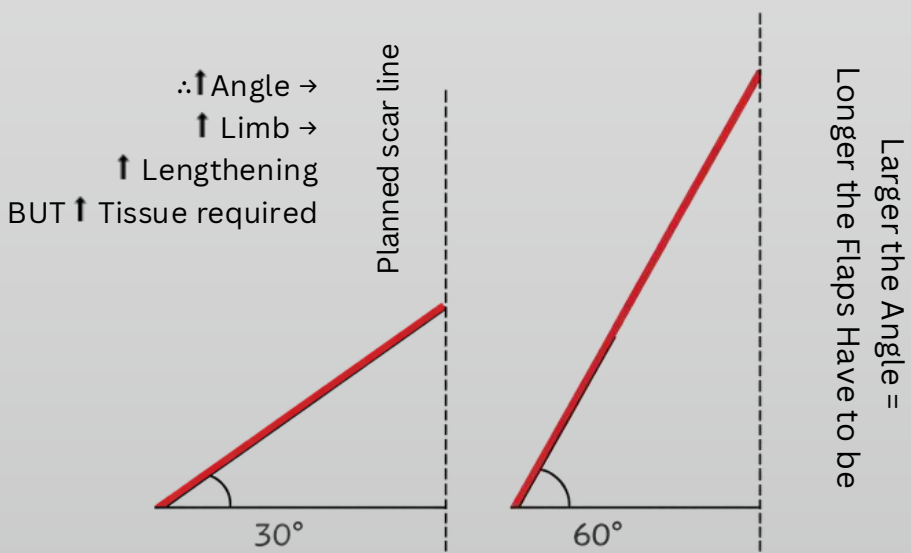
PRINCIPLES | PLAN | ANGLES

PRINCIPLES

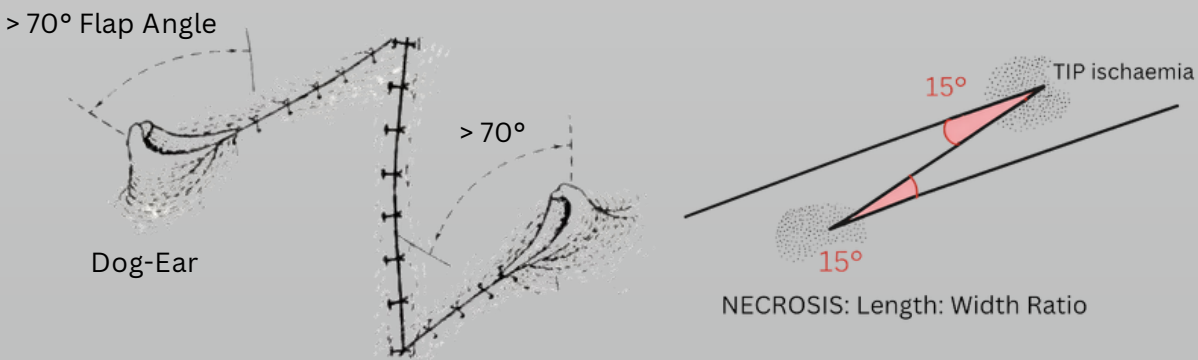
Central limb is usually aligned with the contracture

Angle of limbs determines how much the scar is lengthened

• Larger angle = more lengthening of the new flap

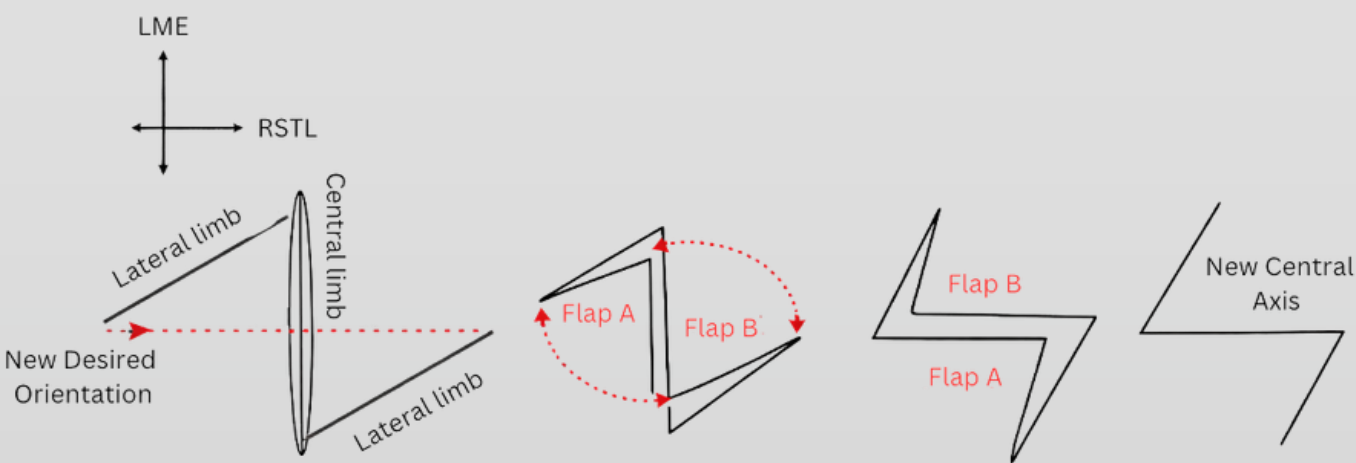
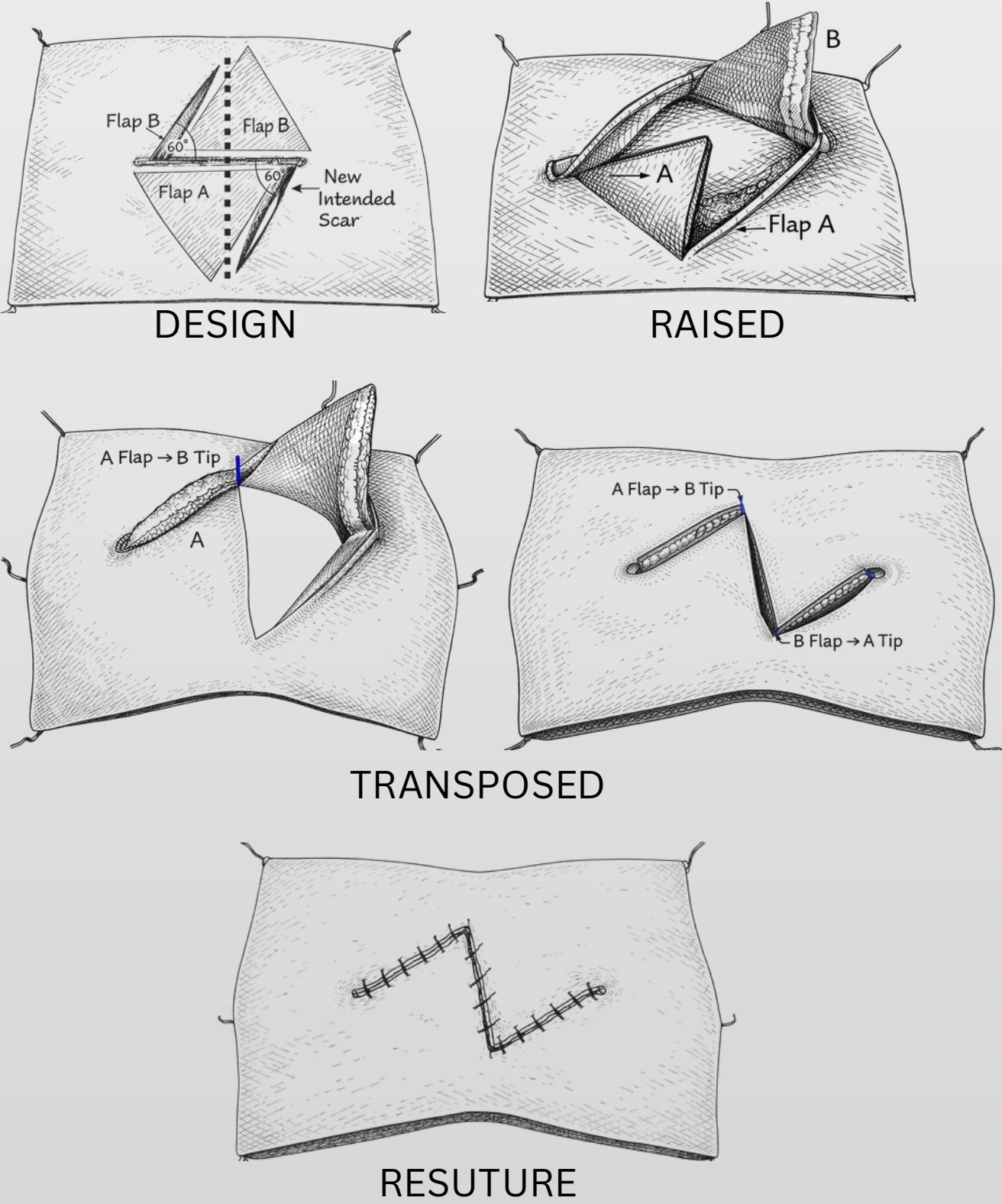


- Common angles: 30°, 45°, 60°
 - > 70° causes dog ear deformity
 - < 30° may cause flap necrosis



Length of central limb – increasing it enlarges the interposed skin flaps, requiring more skin laxity

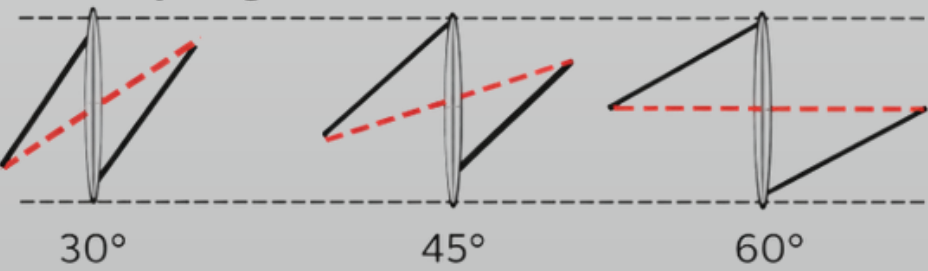
PLAN



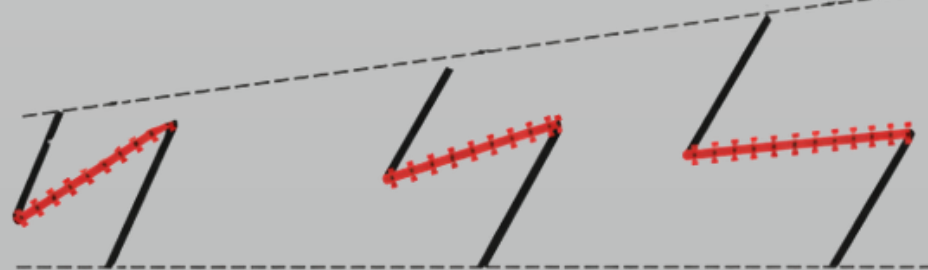
ANGLES V. SCAR LENGTH

30°	25%
45°	50%
60°	75%
75°	100%

Z-Plasty Angle



25% 50% 75%

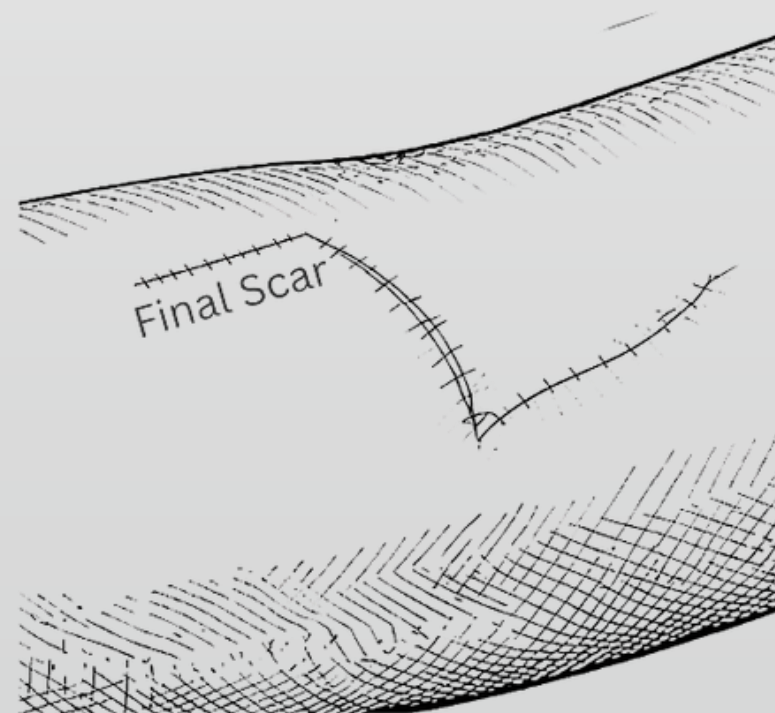
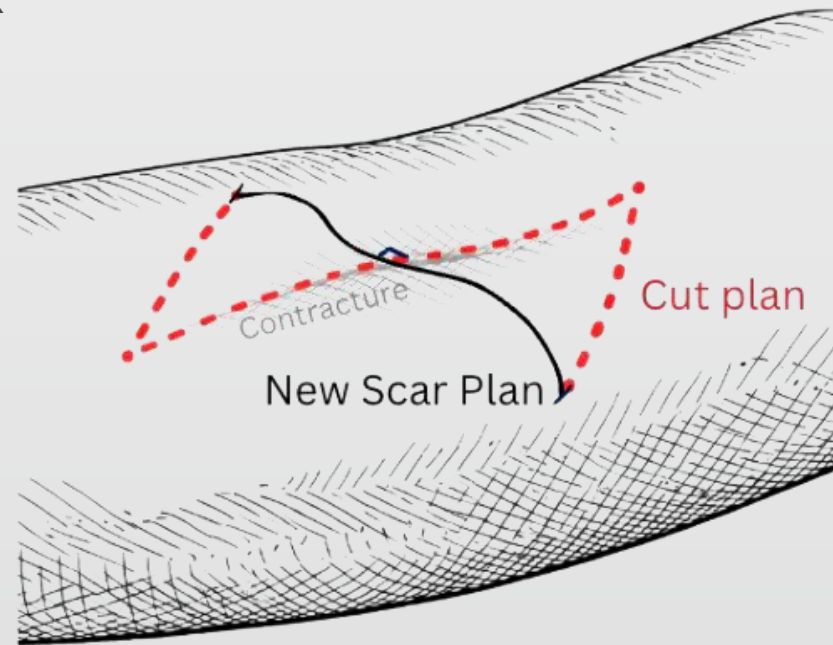


Length Gain

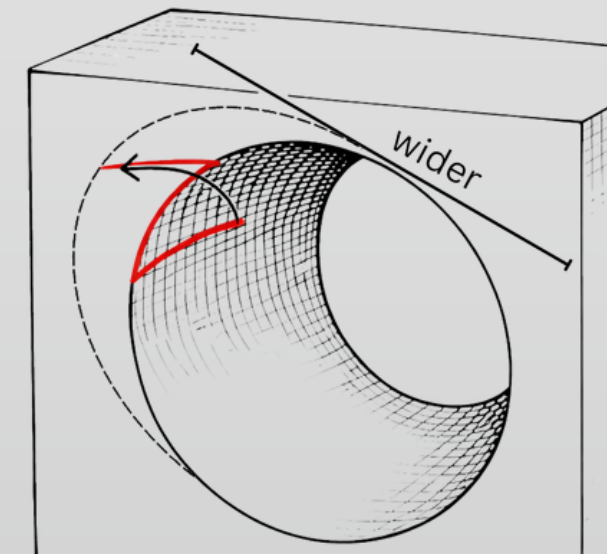
TYPES

LINEAR | CIRCULAR

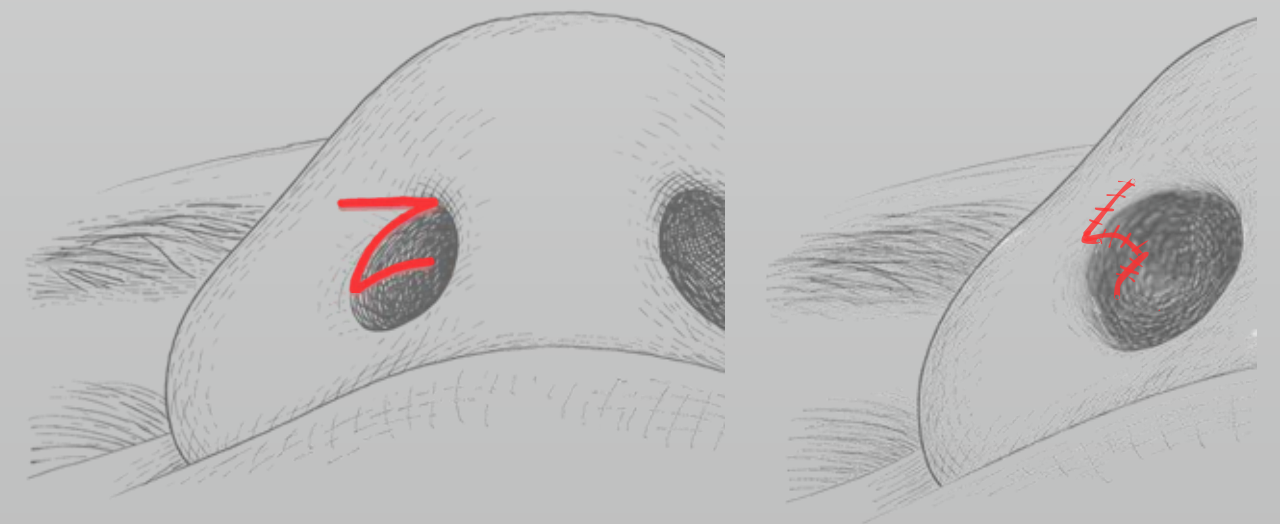
LINEAR



CIRCULAR



E.G. AXIAL PLANE VIEW OF THE NOSE



TECHNIQUE

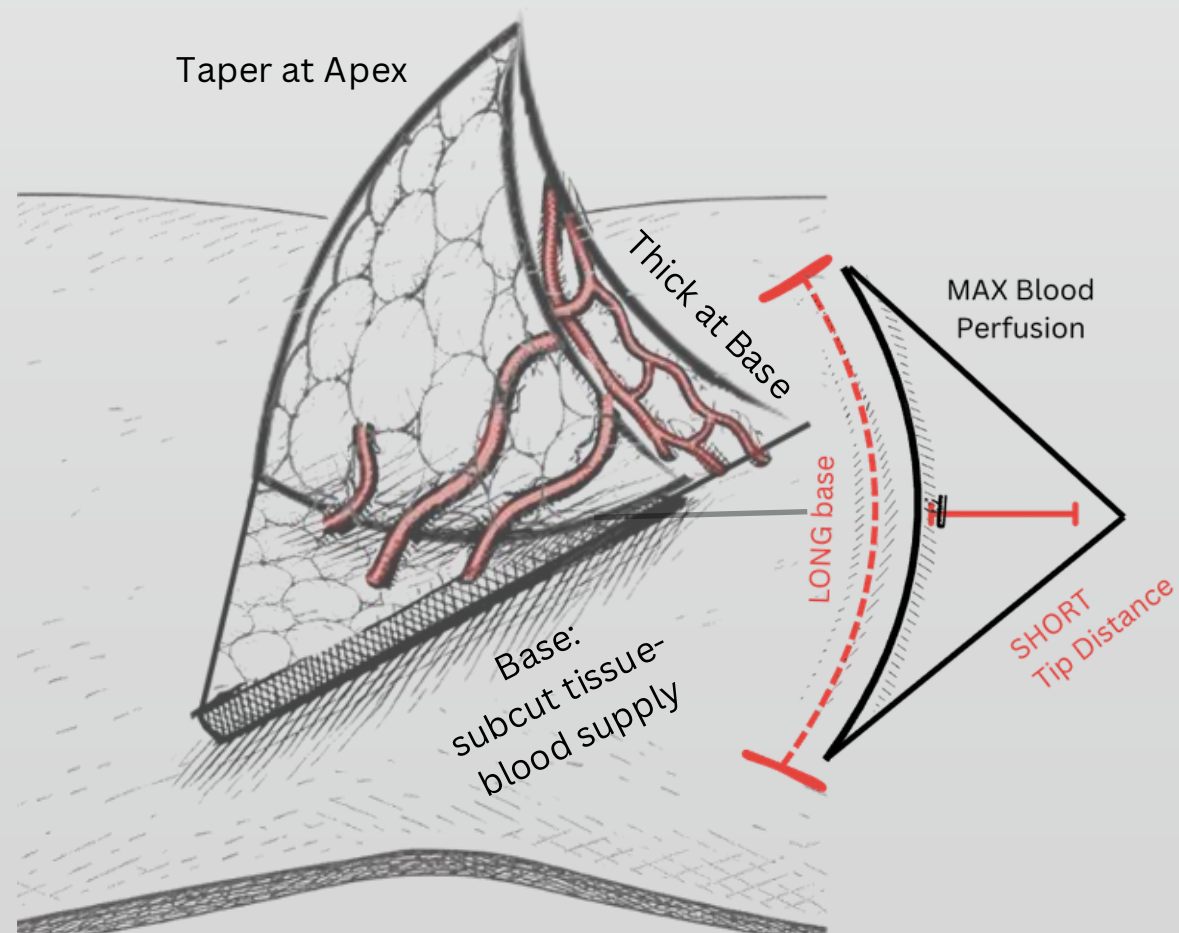
STEPS | OVERVIEW

STEPS

1. Mark the incision

2. Incise through epidermis and dermis (in the subcutaneous plane between muscle and dermis)

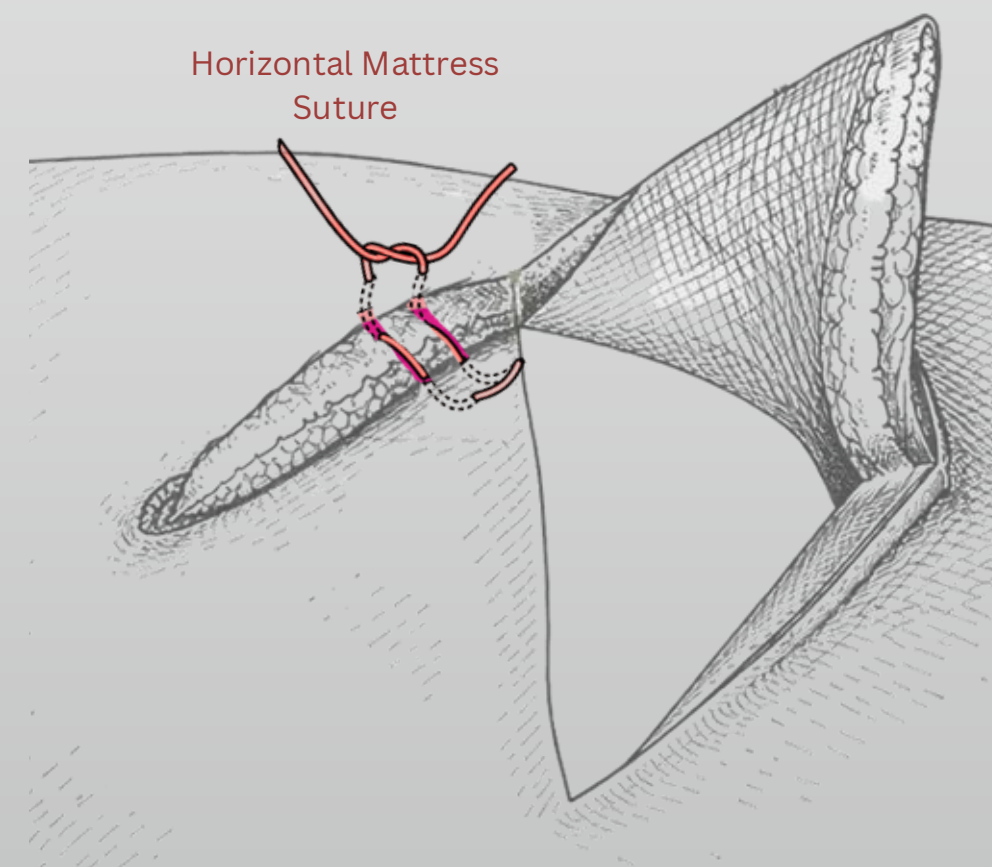
Tip: Include a min of 4mm of subcutaneous tissue in your flaps to maximise viability.



3. Lift the triangle flaps

4. Transpose flaps – reorient each apex to the far corner of the opposite flap

- Use *buried horizontal mattress sutures*.

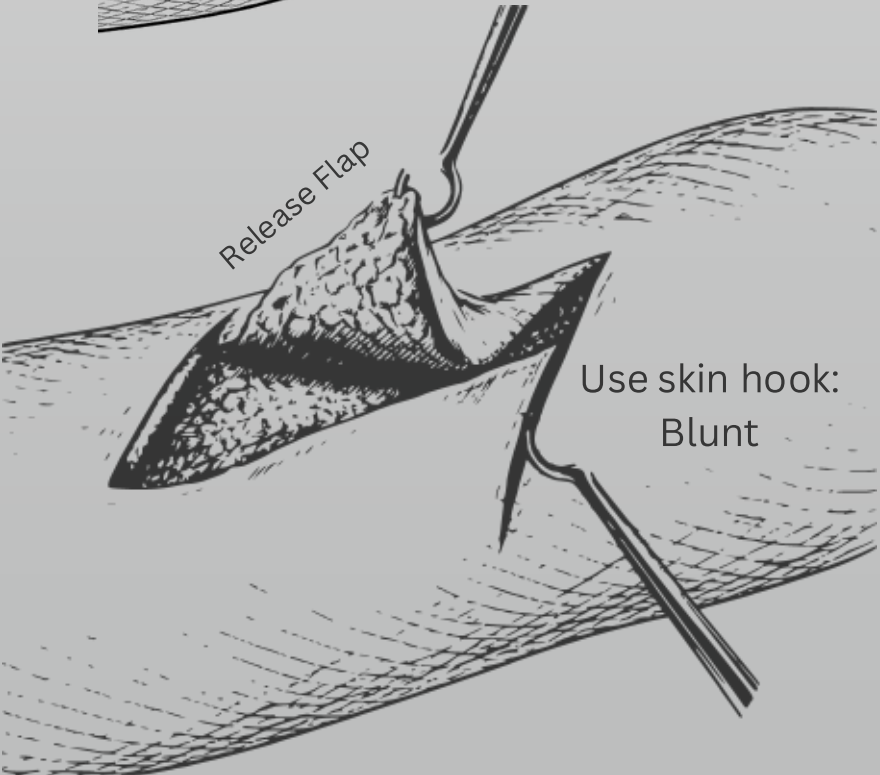
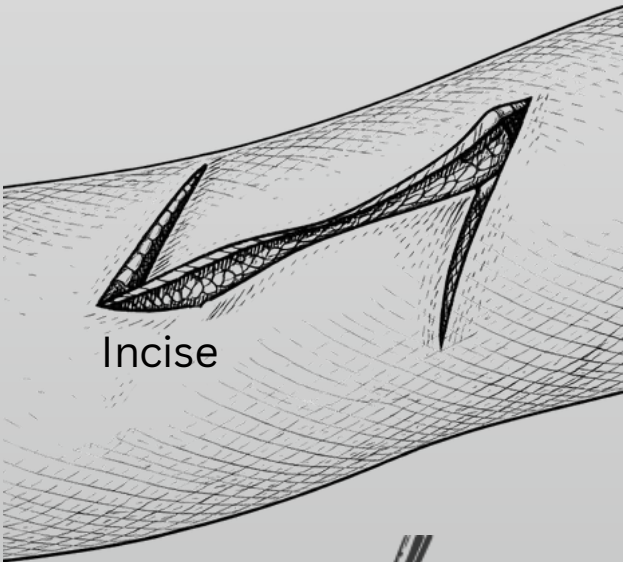
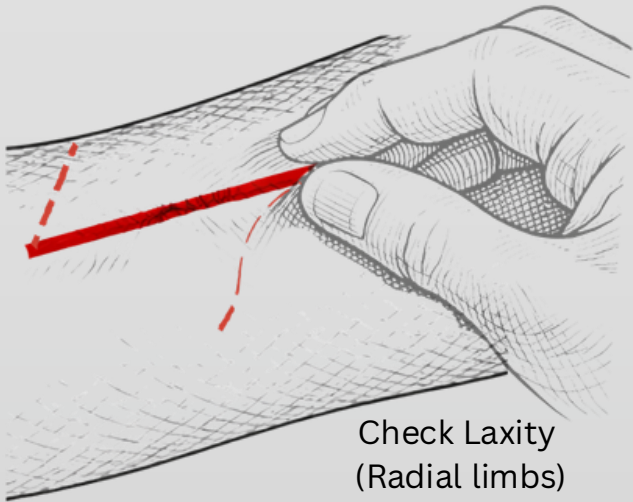
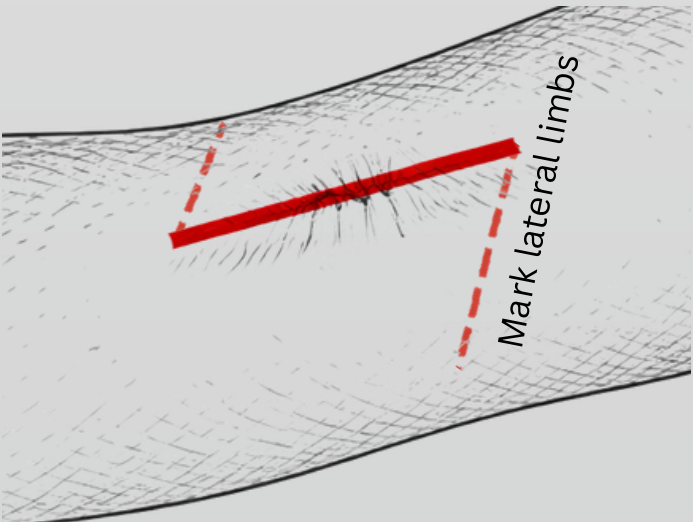
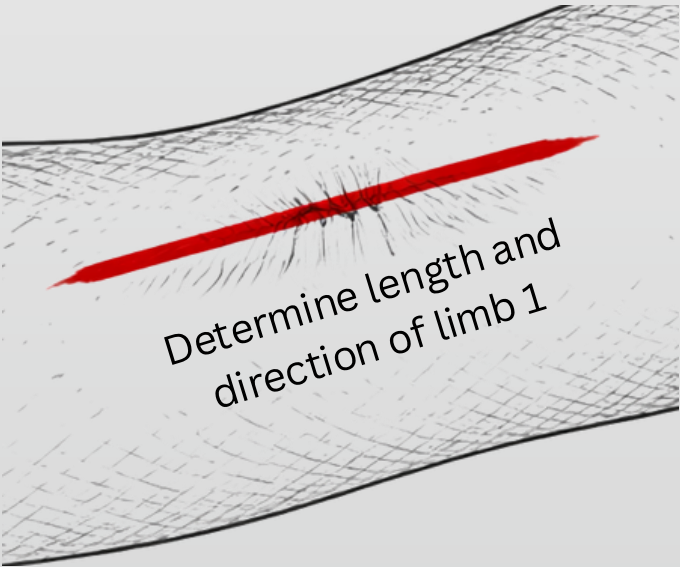
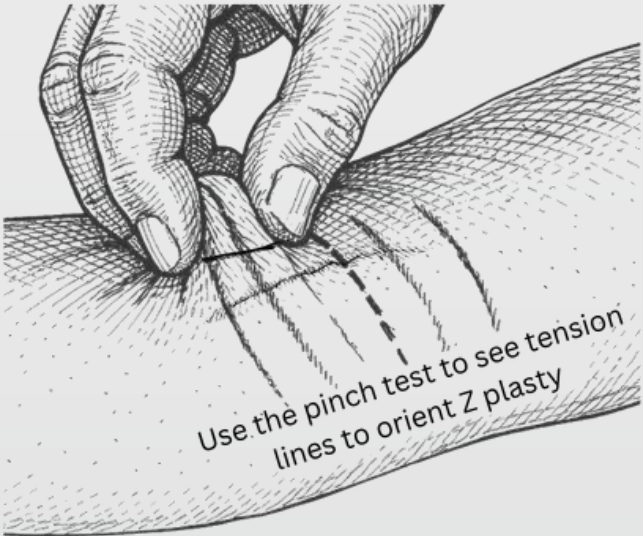


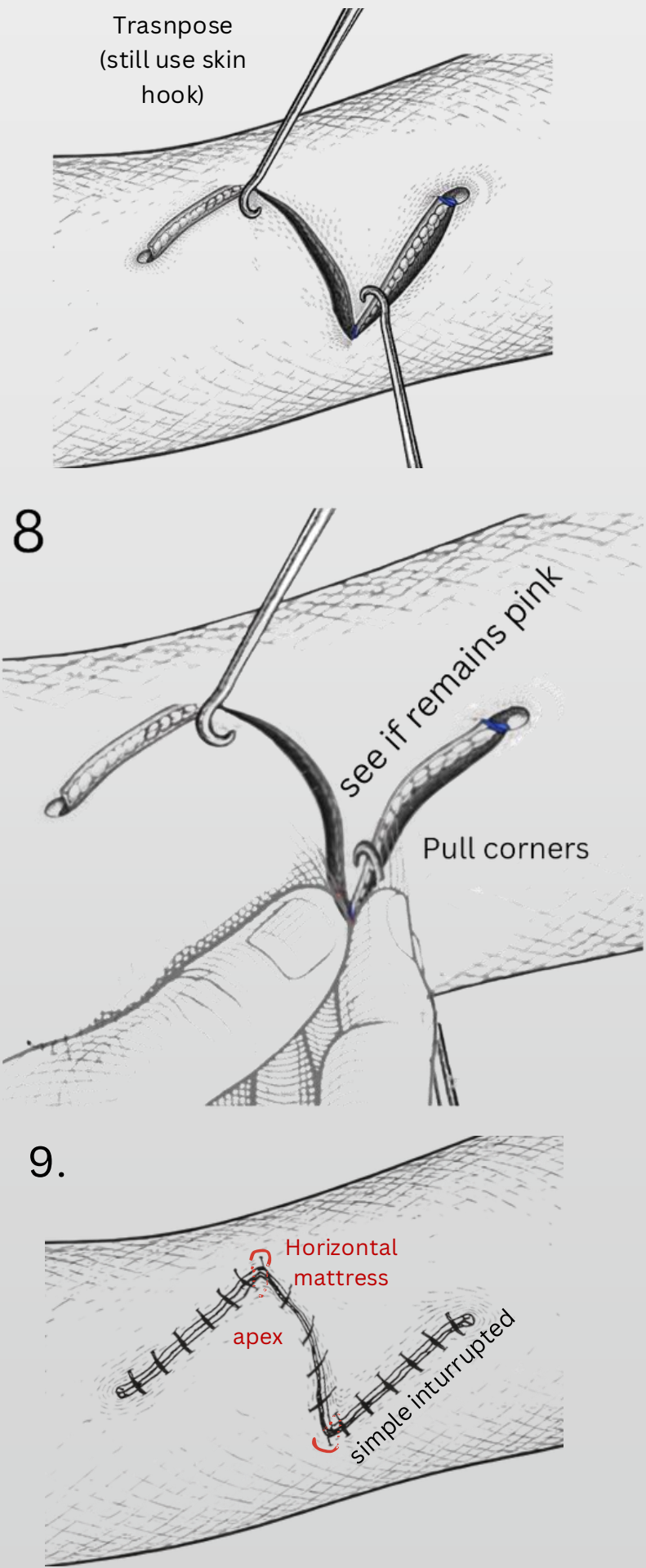
5. Check tension – if too much tension, release more of the flap and try again

6. Confirm viability – check for skin edge bleeding

OVERVIEW

1.

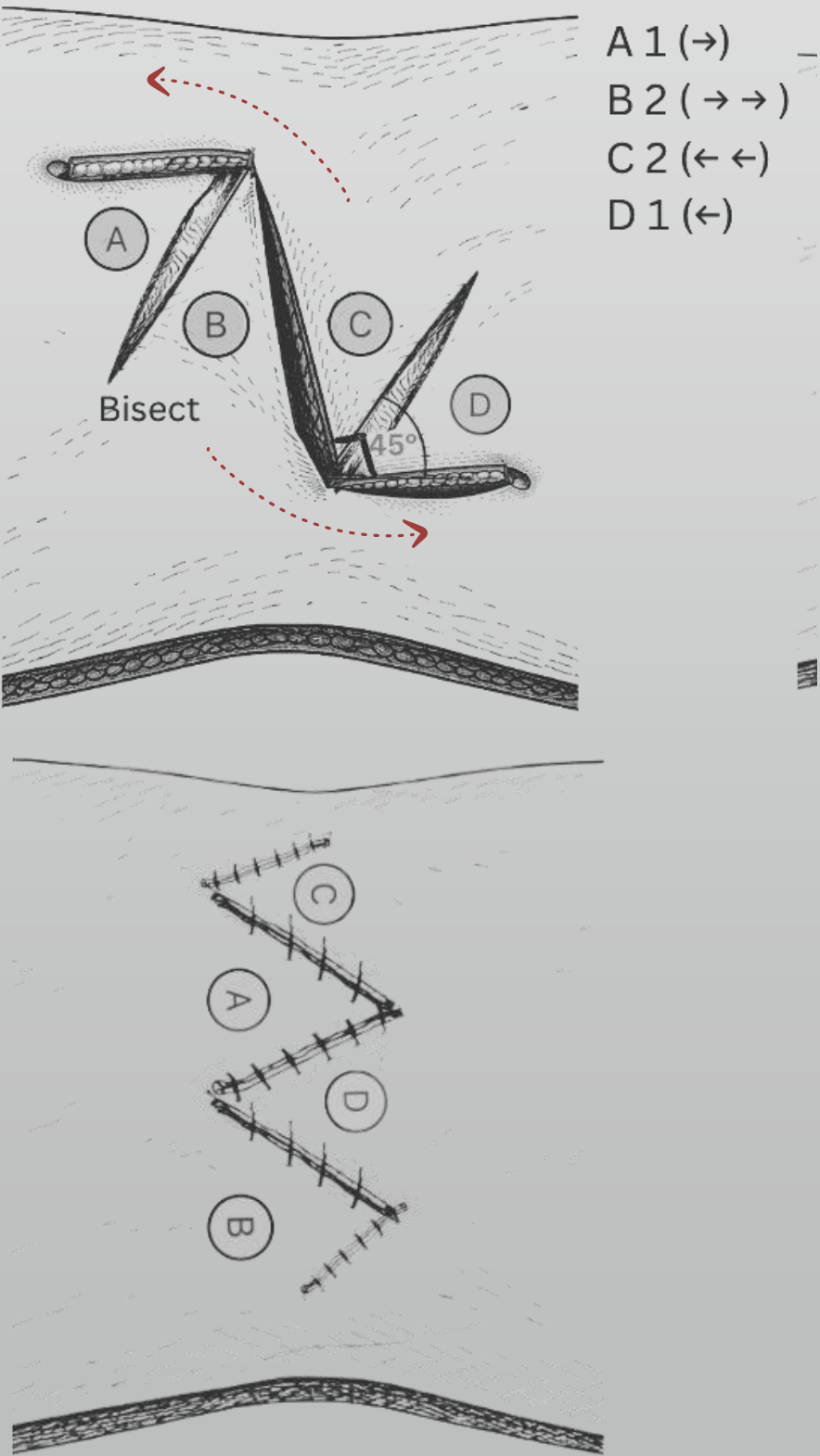




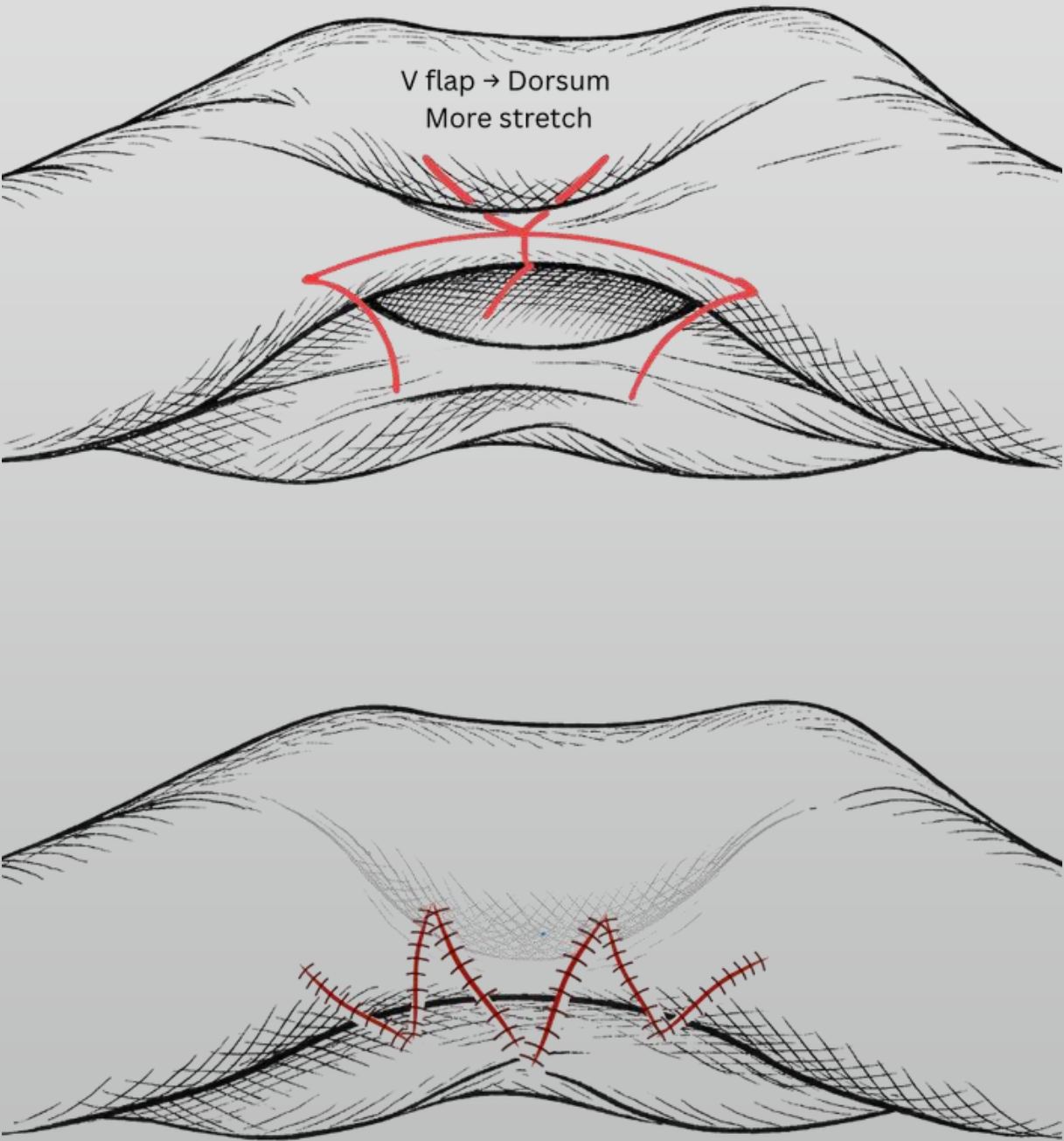
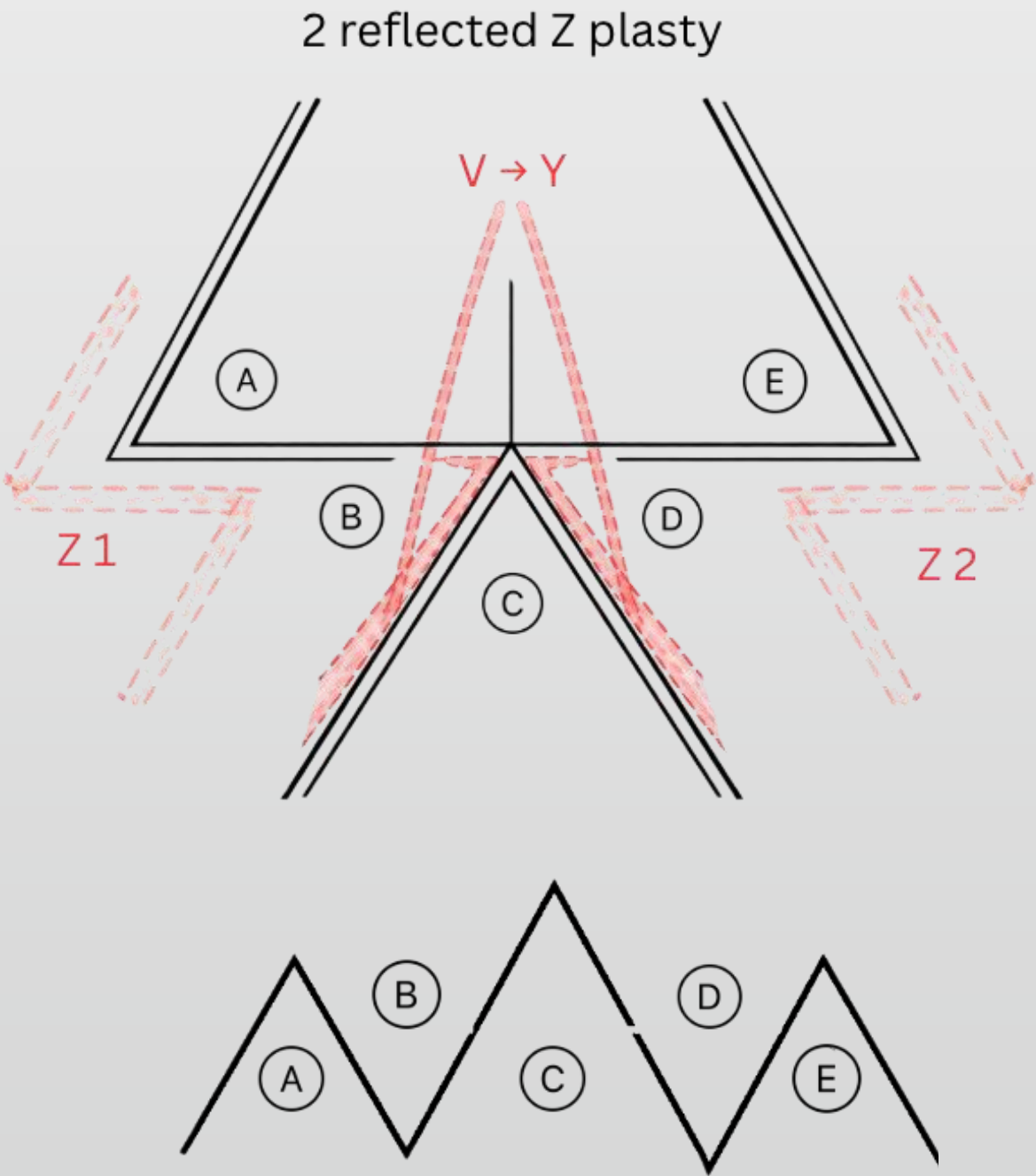
OTHER FLAPS

4 FLP / JUMPING MAN \ STACKING

4- FLAP DESIGN



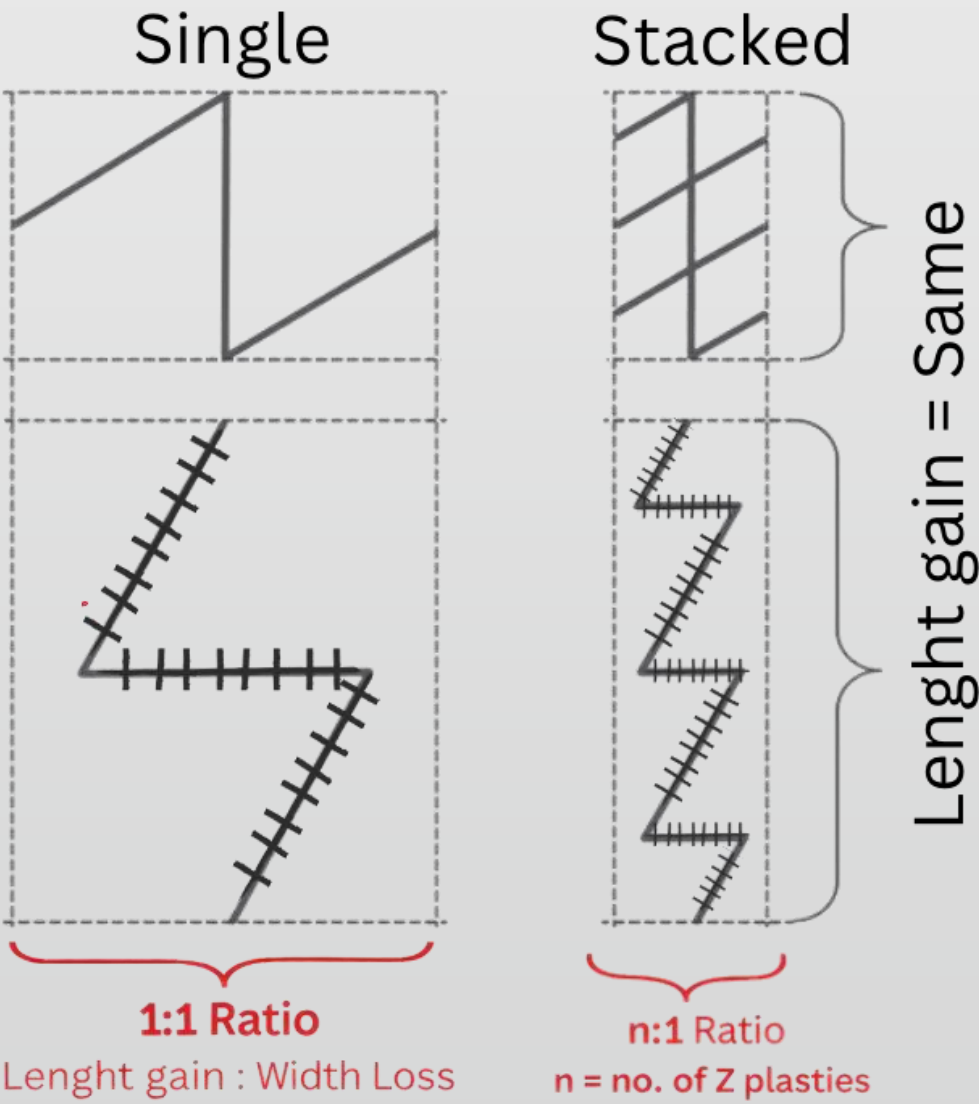
JUMPING MAN



STACKING

Used when there is limited adjacent tissue or in sensitive areas (face, hand)

Multiple small Z-plasties can be stacked to achieve the effect of one large Z-plasty



ADVERSITIES

CONTRAINDICATIONS \ COMPLICATIONS

CONTRAINDICATIONS

- Inadequate tissue nearby
- Aesthetic landmarks nearby
- Poor skin laxity
- Risk of necrosis (if angle too small)

COMPLICATIONS

- **Necrosis** – flap too thin
- **Flap damage** – rough handling
- **Dehiscence** – splitting/bursting open
- **Infection**

ENDNOTE

Acknowledgements

- This guide was developed primarily from content and teachings provided by the SurgHub Z-Plasty Course (www.surghub.org/course/z-plasty), which served as the principal educational resource and framework for this document.

Illustrations

- The surgical illustrations featured throughout this guide were produced using a mixed-media approach. Each figure was originally hand-drawn/ digitally designed and then enhanced using DALL-E 3 by OpenAI, an advanced generative image-synthesis model. This process enabled the preservation of anatomical accuracy from the original sketches while achieving a clean, publication-quality finish.

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